

THE READOUT, NOT THE TARGET: MULTI-RESOLUTION RELATIONAL DISTILLATION FOR COMPACT TEXT EMBEDDINGS

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Paper under double-blind review

ABSTRACT

Knowledge distillation for text embedding models must transfer the teacher’s geometry: which examples are neighbours, and whether a similarity score means the same thing for one query as for another. The dominant family of methods does this relationally, asking the student to match the teacher’s softmax distribution over a set of candidate neighbours, and recent work improves them by enriching that supervision — multiple teachers, views, layers, or multiple resolutions of a teacher-induced neighbourhood graph. We show that these methods all read the student out the same way, through a *single* softmax over the candidate set, and that this shared choice, rather than any property of the target, causes three failures no target can repair. Matching several targets through one readout is algebraically equal to matching their *mixture* alone plus a constant, so the additional targets change nothing and the attained minimum is a divergence the targets fix before training begins. A single softmax determines the student’s cosine profile only up to a per-anchor additive constant, which is why relational distillation transfers within-anchor ranking but leaves unconstrained the absolute similarity levels that globally thresholded retrieval and Spearman correlation depend on. And targets built on a sparsified neighbourhood graph — the standard defence against hubness — place a strictly positive gradient on every unsupported column, so the objective separates exactly the pairs the teacher ranks as near but the graph discarded. We therefore replace the single softmax with a *bank* of softmaxes at distinct temperatures, over a scale family whose zero-step member is the teacher’s own dense kernel before any sparsification. The resulting objective, HeatBank, is a single term with no balancing weight, adds no parameters, and needs only cached teacher embeddings. Distilling LLM-scale embedding teachers into a compact encoder, it improves over the strongest prior relational objective, and the improvement concentrates in the level-sensitive metrics the analysis singles out rather than in the probe-based ones. Because each result names the quantity a single readout leaves free, it predicts which metrics each component should move and which it should leave unchanged, so the ablations can falsify the analysis rather than merely rank configurations.

1 INTRODUCTION

The value of a text embedding model comes less from individual vectors than from the geometry they induce: which examples are neighbours, and whether a similarity score means the same thing for one query as for another. Since strong embedding models are increasingly built on large language-model backbones (Chen et al., 2024; Zhang et al., 2025b), they are expensive to serve, and distilling them into small encoders that preserve that geometry is a practical necessity.

Existing methods differ in the target, not in the readout. The dominant approach is relational: rather than regressing student coordinates onto teacher coordinates, the student is asked to preserve relations among examples (Park et al., 2019; Tung & Mori, 2019), in practice by matching a teacher-induced distribution over candidate neighbours (Fang et al., 2021; Tian et al., 2020; Yang et al., 2020). Progress in this line has come almost entirely from making that distribution carry more:

distilling from several teachers or several stages (Zhang et al., 2024a), adding views and alternative divergences (Xu et al., 2023; Wen et al., 2023), supervising internal states under capacity mismatch (Zhang et al., 2024b; Dao et al., 2026), or — reading the teacher space as a sampled manifold — deriving targets at several resolutions from random walks on a neighbourhood graph (Coifman & Lafon, 2006; Klicpera et al., 2019). Whatever the target, the student side is left untouched: a cosine profile over the candidate set is converted to a distribution by *one* softmax at *one* temperature, and the several targets are matched against that same distribution through a weighted sum of KL divergences. Our starting point is that this uniform, unexamined choice is where these methods lose the geometry they set out to transfer.

Three failures of the shared readout. None of the following is a statement about targets; each holds for any relational objective of the form above. First, cross-entropy is linear in its target, so a weighted sum of KL divergences against *one* student distribution collapses onto the mixture of the targets plus a target-only constant (Proposition 1). Enriching the target family therefore buys smoothing and nothing else — two families with the same mixture induce identical gradients — and the attained minimum is a generalised Jensen–Shannon divergence among the targets, a number determined before any training (Corollary 1), so a loss curve that flattens is not evidence of convergence. This applies verbatim to the multi-teacher, multi-view and multi-layer variants above. Second, a softmax is invariant to a constant shift of its logits, so a single readout pins the student’s cosine profile only up to a per-anchor additive constant (Proposition 2): within-anchor ranking is transferred, absolute similarity *levels* are not. This is the precise reason relational distillation is observed to preserve neighbour order while degrading tasks that apply one global threshold or rank all pairs jointly, and it is what the usual remedy — a pointwise term aligning s_i to t_i through a learned map — provably cannot fix, since its invariance group is strictly larger (Lemma 1). Third, targets derived from a sparsified graph assign mass zero outside the anchor’s support, where the cross-entropy gradient is strictly positive (Proposition 3); the objective thus pushes apart precisely those pairs the teacher scores as near but the mutual-neighbour test discarded — so the standard defence against hubness (Radovanović et al., 2010) creates a second, compounding failure.

One change addresses all three. Each failure is a statement about how the student reads a similarity vector out as a distribution, so we intervene there. We replace the single softmax with a bank of softmaxes at distinct temperatures, one per scale. Distinct temperatures break the algebraic identity behind Proposition 1, and they over-determine the cosine profile, so the minimiser depends on similarity levels rather than only on their shift-equivalence class (Proposition 2). For the support pathology we add no auxiliary term but extend the scale family downwards: the zero-step member is the teacher’s own dense kernel, taken *before* sparsification, and is the unique member whose support is the full candidate set. The family thus interpolates between two extremes of one construction — no graph at $r = 0$, deep walks at large r — rather than gluing a relational loss to a diffusion loss. The result, HeatBank, is a one-term objective with no balancing weight, requiring only cached teacher embeddings: no hidden states, no online teacher pass, no architecture change, no shared tokenizer.

Our contributions are the three results above — a collapse identity with a pre-training floor (Section 4.1), an under-determination result and its removal by over-determination (Section 4.2), and a support result with the zero-step repair (Section 4.3) — together with HeatBank, the objective they force, and build-time diagnostics that make each failure checkable before training (Section 5). Because each proposition names the quantity left unconstrained, it also says which downstream metrics should move when the corresponding component is removed and which should not. We report results split along that line — metrics defined by a single global decision rule on cosine against metrics that use only within-anchor ordering — so the ablations test the analysis rather than merely rank configurations.

2 RELATED WORK

Compact text embedding models and distillation. Sentence-BERT made transformer encoders practical for sentence-level similarity (Reimers & Gurevych, 2019), contrastive objectives such as SimCSE showed strong sentence embeddings can be learned with minimal supervision (Gao et al., 2021b), and broad benchmarks shifted evaluation to retrieval, classification and clustering suites (Thakur et al., 2021; Muennighoff et al., 2023; Chung et al., 2025). Recent systems scale the back-

bone and training mixture (Chen et al., 2024; Zhang et al., 2025b;a), which makes them strong teachers but expensive to serve; we treat them strictly as black-box teachers, using only their final vectors. Distillation of pretrained encoders transfers through logits, hidden states and attention relations (Hinton et al., 2015; Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020); recent LLM distillation adds on-policy training and alternative divergences (Agarwal et al., 2024; Ko et al., 2025; Wen et al., 2023; Xu et al., 2024), and cross-tokenizer methods address vocabulary mismatch (Zhang et al., 2024b; Chen et al., 2025). Embedding-specific work distills LLM-generated retrieval data (Lee et al., 2024), uses multi-stage distillation (Zhang et al., 2024a), aligns students for asymmetric retrieval (Vujanic & Ruecksties, 2025), or combines task-targeted distillation with contrastive training (Akram et al., 2026). Every one of these adds a supervision channel and none changes how the student turns a similarity vector into a distribution, so the failures of Section 4 are inherited by all of them: a richer signal read out through one softmax is still constrained only up to a per-anchor shift, and combining several such signals still reduces to their mixture.

Relational and neighbourhood-distribution objectives. Relational knowledge distillation preserves relations among examples rather than coordinates (Park et al., 2019), and similarity-preserving distillation matches mini-batch Gram matrices (Tung & Mori, 2019). Closest to the zero-step member of our scale family is SEED (Fang et al., 2021), in which the student matches the teacher’s softmax similarity distribution over a queue of instances; contrastive distillation (Tian et al., 2020; Gao et al., 2021a; Xu et al., 2023) transfers related pairwise preferences, and in the graph setting local-structure-preserving distillation matches neighbourhood distributions between networks (Yang et al., 2020; Zhang et al., 2022). This is the family our analysis targets, and it has two blind spots. Each of these methods defines the student distribution by a single softmax at a single temperature and studies only what the target should be, so none asks what the readout itself determines; the well-known observation that relational objectives preserve neighbour order but not comparable similarity scales is reported as an empirical limitation rather than derived, and Proposition 2 shows it is forced. And where several targets are used at once — multiple teachers, layers, views or resolutions — they are averaged against that one distribution, which Proposition 1 shows is a single-target objective on the mixture. Our contribution is therefore not another target: HeatBank is the objective these results force. In particular the zero-step target coincides with a SEED-style similarity distribution, which we make explicit precisely so the multi-resolution component is measured *on top of* that baseline rather than confounded with it.

Graph diffusion, manifolds, and internal alignment. Neighbourhood graphs expose low-dimensional structure through spectral coordinates (Belkin & Niyogi, 2003), Markov diffusion (Coifman & Lafon, 2006), and heat diffusion for visualisation and geodesic-preserving embedding (Moon et al., 2019; Hugué et al., 2023); powers of a transition operator are a standard filter bank in graph learning (Klicpera et al., 2019). We borrow this machinery only to *generate* a target family — no spectral coordinates are used as an output and no eigendecomposition is required. The multi-scale motivation carries over from these settings, but the consumption does not: a filter bank is informative when its outputs are read separately, whereas a distillation objective that pushes them through one softmax recovers only their mixture. Two further practices in this line become liabilities under distillation. Mutual k NN construction is the standard defence against hubness (Radovanović et al., 2010), and Section 4.3 shows it turns every discarded near pair into a repulsion; and diffusion beyond a few steps, harmless when the operator is used for visualisation, makes the scales duplicates of one another. Orthogonally, representation-similarity matching (Kornblith et al., 2019), optimal-transport alignment (Alqahtani et al., 2021) and teacher-anchored layer alignment (Dao et al., 2026) supervise internal states under capacity mismatch; these need teacher internals, which a black-box embedding API does not expose, and they constrain the student’s coordinates rather than the similarity levels our analysis isolates. We include the strongest as baselines.

3 SETUP: A SCALE FAMILY OF TEACHER TARGETS

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be a training corpus, T a frozen teacher encoder producing $t_i = T(x_i) \in \mathbb{R}^{d_T}$, and S_θ a compact student producing $s_i = S_\theta(x_i) \in \mathbb{R}^{d_S}$. Teacher embeddings are computed once, cached, and the teacher discarded; only cosine similarities of teacher vectors are ever used, so nothing is assumed about d_T , the teacher’s tokenizer, or its internal states.

Teacher affinity. For a temperature $\tau > 0$ define the affinity kernel

$$K_{ij}^{(\tau)} = \exp(\cos(t_i, t_j)/\tau) \mathbf{1}_{i \neq j}, \quad (1)$$

which is dense: every ordered pair receives positive affinity. A neighbourhood graph is obtained by masking K to mutual k -nearest neighbours,

$$M_{ij} = \mathbf{1}[j \in \mathcal{N}_k(i) \wedge i \in \mathcal{N}_k(j)], \quad W = M \odot K^{(\tau_g)}, \quad (2)$$

where $\mathcal{N}_k(i)$ is the teacher’s top- k neighbour set of i . Mutuality is the standard defence against hubness (Radovanović et al., 2010); Section 4.3 shows it has a cost. With $D = \text{diag}(W\mathbf{1})$ we write $P = D^{-1}W$ for the row-stochastic transition matrix and $Q = \frac{1}{2}(I + P)$ for its lazy variant.

The scale family. Sparsification and diffusion are two separate operations on the same kernel, and indexing them jointly yields a single family. Let $\bar{\mathcal{R}} = \{0\} \cup \mathcal{R}$ with $\mathcal{R} \subset \mathbb{N}_{>0}$, and define

$$P_r = \begin{cases} \text{rownorm}(K^{(\tau_0)}) , & r = 0 \quad (\text{no sparsification, no diffusion}), \\ Q^r, & r \in \mathcal{R} \quad (r\text{-step lazy walk on the sparsified kernel}). \end{cases} \quad (3)$$

The index r therefore reads as *how much graph structure has been imposed*. At $r = 0$ the teacher is asked directly: row i of P_0 is its own similarity distribution over the whole corpus. At $r \geq 1$ the kernel is first pruned to a manifold approximation, then mass is propagated along it, so larger r describes broader semantic regions. The two extremes are not different kinds of supervision but the endpoints of one construction, which is why HeatBank needs no second loss term and no balancing weight.

Laziness is what makes $\mathcal{R}$ graded rather than a set of near-duplicates. Since $\text{spec}(P) \subseteq [-1, 1]$, the map $\lambda \mapsto \frac{1}{2}(1 + \lambda)$ sends $\text{spec}(Q)$ into $[0, 1]$, removing sign oscillation and roughly doubling the mixing time. Scales stay distinguishable only while $|\lambda_2(Q)|^r$ has not decayed, which for a plain walk on a mutual k NN graph fails within about two steps.

Candidate sets and restricted targets. Scoring an anchor against all $N - 1$ columns is infeasible, so each anchor i is scored against a candidate set $C_i \subset [N] \setminus \{i\}$ with $|C_i| = C \ll N$. The target at scale r is the corresponding row of P_r restricted and renormalised,

$$p_{i,r}(j) = \frac{(P_r)_{ij}}{\sum_{j' \in C_i} (P_r)_{ij'}}, \quad j \in C_i. \quad (4)$$

For $r \geq 1$ we additionally retain only a sparse pool $\mathcal{P}_i$ of size P per anchor, whose truncation residual $\varepsilon_{i,r} = 1 - \sum_{j \in \mathcal{P}_i} (P_r)_{ij}$ is reported rather than assumed negligible. Note $\text{supp}(p_{i,0}) = C_i$ while $\text{supp}(p_{i,r}) \subseteq \mathcal{P}_i \cap C_i$ for $r \geq 1$; this asymmetry is the subject of Section 4.3.

Student readout. Writing $v_i \in \mathbb{R}^C$ for the student’s cosine profile, $v_i(j) = \cos(s_i, s_j)$, a readout at temperature τ is

$$q_i^\tau(j) = \frac{\exp(v_i(j)/\tau)}{\sum_{j' \in C_i} \exp(v_i(j')/\tau)}. \quad (5)$$

Every objective below is a weighted sum of $D_{\text{KL}}(p_{i,r} \| q_i^{\tau_r})$ over $r \in \bar{\mathcal{R}}$ with $\omega_r \geq 0$, $\sum_r \omega_r = 1$. The only design question is how many distinct temperatures the readout uses, and Section 4 shows that this single question decides whether the objective is multi-resolution at all, whether it constrains similarity levels, and whether it agrees with the teacher off the graph.

4 THREE FAILURES OF A SINGLE READOUT

We now state what goes wrong when the objective of Section 3 is optimised through one softmax, and what each failure forces. All three results are statements about the readout equation 5; none depends on the targets being diffusion distributions.

4.1 A SINGLE READOUT MAKES MULTIPLE SCALES VACUOUS

Proposition 1 (Collapse). *Fix an anchor i , targets $\{p_r\}_{r \in \bar{\mathcal{R}}}$ on C_i , and weights $\omega_r \geq 0$ with $\sum_r \omega_r = 1$. Let $\bar{p} = \sum_r \omega_r p_r$. Then for any single distribution q on C_i ,*

$$\sum_{r \in \bar{\mathcal{R}}} \omega_r D_{\text{KL}}(p_r \| q) = \underbrace{H(\bar{p}, q)}_{\text{depends on } \bar{\mathcal{R}} \text{ only via } \bar{p}} - \underbrace{\sum_{r \in \bar{\mathcal{R}}} \omega_r H(p_r)}_{\text{independent of } q}, \quad (6)$$

where $H(\cdot, \cdot)$ is cross-entropy and $H(\cdot)$ is entropy. Consequently ∇_q of the left-hand side depends on the scale family only through $\bar{p}$: any two families with the same mixture induce identical optimisation problems.

The proof is one line: $D_{\text{KL}}(p_r \| q) = -H(p_r) + H(p_r, q)$, and $H(\cdot, q)$ is linear in its first argument. The identity is elementary, its consequence less so: an objective built to transfer structure at several resolutions transfers, up to a constant, exactly what a single-resolution objective on the smoothed target $\bar{p}$ transfers, so choosing $\bar{\mathcal{R}} = \{1, 2, 4\}$ over $\{1\}$ buys target smoothing and nothing else. This holds verbatim for multi-teacher, multi-view and multi-layer distillation.

Corollary 1 (Irreducible floor). *The left-hand side of equation 6 is minimised at $q = \bar{p}$, where it attains*

$$\text{JS}_\omega(\{p_r\}_{r \in \bar{\mathcal{R}}}) = H(\bar{p}) - \sum_{r \in \bar{\mathcal{R}}} \omega_r H(p_r) \geq 0, \quad (7)$$

the generalised Jensen–Shannon divergence of the family. This quantity is a function of the teacher targets alone and can be computed before the first gradient step.

Corollary 1 makes a tied-temperature loss curve uninterpretable: a plateau may indicate a converged student or an exhausted objective, and the two have opposite remedies. We therefore compute equation 7 offline and report the excess $\mathcal{L} - \text{JS}_\omega$; a near-zero JS_ω certifies that the scales carry the same target.

What this forces. If the objective is to depend on $\bar{\mathcal{R}}$ beyond its mixture, the student side must vary with r . Assigning scale r its own readout temperature τ_r is the minimal such change: no parameters, no projection, no extra forward cost.

4.2 A SINGLE READOUT LEAVES SIMILARITY LEVELS FREE

Proposition 2 (Under- and over-determination). *Let $v_i \in \mathbb{R}^C$ be the student’s cosine profile and q_i^τ the readout equation 5.*

- (a) *For a single temperature τ , q_i^τ is invariant under $v_i \mapsto v_i + a\mathbf{1}$ for every $a \in \mathbb{R}$. Any objective depending on v_i only through q_i^τ therefore determines v_i at best up to a per-anchor additive constant.*
- (b) *For $|\bar{\mathcal{R}}| \geq 2$ with distinct temperatures, exact simultaneous satisfaction $q_i^{\tau_r} = p_{i,r}$ for all r requires*

$$\tau_r \log p_{i,r} - \tau_{r'} \log p_{i,r'} \in \text{span}\{\mathbf{1}\} \quad \text{for all } r, r' \in \bar{\mathcal{R}}, \quad (8)$$

which fails for generic targets. The system is over-determined, and the minimiser depends on the level of v_i , not only on its shift-equivalence class.

The proof is in Appendix A. Part (a) is why relational distillation transfers ranking but not calibration: the student may offset all similarities of an anchor without changing the loss, so a cosine of 0.8 for one anchor need not mean what 0.8 means for another. Part (b) shows that a bank of at least two distinct temperatures removes exactly this freedom, without any comparison between the student’s metric and the teacher’s.

Prediction 1. Tying the readout temperatures should degrade metrics defined by a single global decision rule on cosine — Spearman correlation over all pairs, pair classification at one global threshold — substantially more than metrics using only within-anchor ordering or fitting a classifier on the embeddings, such as linear-probe F_1 .

This split is what makes Proposition 2 falsifiable: an average improvement is consistent with many explanations, a component that moves level-sensitive metrics while leaving probe-based ones unchanged with few.

4.3 SPARSE TARGETS ANTI-LEARN OFF THE GRAPH

Proposition 3 (Support pathology). *Let p be a target on C_i with $\text{supp}(p) \subsetneq C_i$, and let $q = q_i^\tau$. For any $j \notin \text{supp}(p)$,*

$$\frac{\partial}{\partial v_i(j)} D_{\text{KL}}(p \| q) = \frac{q(j) - p(j)}{\tau} = \frac{q(j)}{\tau} > 0. \quad (9)$$

Gradient descent therefore decreases $\cos(s_i, s_j)$ monotonically, at a rate independent of $\cos(t_i, t_j)$.

The proof is immediate: with $\ell = v_i/\tau$, $\partial D_{\text{KL}}(p \| q)/\partial \ell_j = q(j) - p(j)$ and $p(j) = 0$ off the support. A zero target is thus not an abstention but an instruction to separate, and for $r \geq 1$ the support lies inside $\mathcal{P}_i$, excluding every j whose teacher similarity to i is high but which failed the mutual-neighbour test of equation 2 — a set hubness makes both large and biased towards genuinely near pairs (Radovanović et al., 2010). A graph-only objective is thus trained to push apart pairs its own teacher calls similar, damaging exactly the calibration Proposition 2 identifies. The two failures compound.

What this forces. A target with full support on C_i is required, and the family equation 3 already contains one: $p_{i,0}$ is a row of the *unsparisified* kernel, so $\text{supp}(p_{i,0}) = C_i$ and every candidate receives its true teacher mass. The zero-step scale is not a regulariser bolted onto a diffusion objective; it is the member of the family at which no graph has yet been imposed.

Prediction 2. Removing $r = 0$ should raise the fraction of teacher-near, off-graph candidates whose student cosine decreases during training towards 1, and the resulting damage should again concentrate in level-sensitive metrics.

4.4 WHY NOT A POINTWISE ANCHOR

The standard response to “relational losses do not fix the space” is a pointwise term $\mathcal{L}_{\text{anc}} = \frac{1}{|B|} \sum_i (1 - \cos(W_a s_i, t_i))$ with a freely learned $W_a \in \mathbb{R}^{d_T \times d_S}$. It cannot do the job. A relational loss depending on the student only through $\{\cos(s_i, s_j)\}$ is invariant under $G_{\text{rel}} = \{s \mapsto cRs : c > 0, R \in O(d_S)\}$, whereas $s \mapsto As$, $W_a \mapsto W_a A^{-1}$ leaves $\mathcal{L}_{\text{anc}}$ invariant for every $A \in GL(d_S)$; hence $G_{\text{anc}} \supsetneq G_{\text{rel}}$ (Lemma 1, Appendix A). Because the anchor’s invariance group strictly contains the relational term’s, adding it cannot constrain any degree of freedom the relational term leaves free — including the anisotropic rescalings and shears one might hope it would fix. Its optimum is the best linear predictor of t_i from s_i , fitted early and then nearly constant. Calibration must come from elsewhere, and by Proposition 2(b) the temperature bank supplies it within the student’s own metric.

Prediction 3. Re-introducing $\mathcal{L}_{\text{anc}}$ should leave downstream metrics unchanged, and the norm of its gradient contribution should decay towards zero within roughly one epoch.

5 HEATBANK

Sections 4.1–4.4 leave one objective standing. Assign each scale $r \in \bar{\mathcal{R}}$ its own readout temperature τ_r , keep the zero-step member of the family so that one target has full support, and add nothing else:

$$\mathcal{L}_{\text{HeatBank}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \bar{\mathcal{R}}} \omega_r D_{\text{KL}}(p_{i,r} \| q_i^{\tau_r}), \quad \bar{\mathcal{R}} = \{0\} \cup \mathcal{R}, \quad (10)$$

with $p_{i,r}$ from equation 4, $q_i^{\tau_r}$ from equation 5, and τ_r increasing over $r \in \mathcal{R}$ so that broader targets are read out more smoothly. There is one term and no balancing weight: the quantity a pointwise anchor was meant to supply is produced by the bank itself (Lemma 1), and the quantity a separate relational term was meant to supply is the $r = 0$ member of the family. Constants, the choice of τ_0 , and the full procedure are in Appendix B.

The scales share the underlying vector v_i but not the distribution built from it. The identity equation 6 therefore no longer applies, and equation 7 becomes a lower bound rather than an attained value — so a run of equation 10 whose loss falls below JS_ω is direct evidence that the bank is doing something a single readout cannot.

Shared columns. Anchors in a batch are scored against the union of all candidates in the batch, $\mathcal{C}_B = \bigcup_{i \in B} \mathcal{C}_i$, with $p_{i,r}(j) = 0$ for $j \in \mathcal{C}_B \setminus \mathcal{C}_i$ and $q_i^{\tau_r}$ masked at $j = i$. Its purpose is not more negatives but *shared* ones: Proposition 2(b) removes the additive freedom within an anchor, and shared columns extend the comparison across anchors, which is the regime level-sensitive metrics measure. Because the candidate texts are encoded regardless, this is free in forward cost; but it also multiplies the number of zero-target columns, so it amplifies Proposition 3 and is compatible with equation 10 only because $r = 0$ is present.

Sampling can reintroduce the collapse. Scoring an anchor against all $N - 1$ columns is infeasible, so $\mathcal{C}_i$ is drawn from some μ_i and redrawn each epoch. The analysis requires one property of μ_i .

Condition 1 (Per-scale coverage). $\Pr_{\mu_i}[\text{supp}(p_{i,r}) \cap \mathcal{C}_i \neq \emptyset] = 1$ for every $r \in \bar{\mathcal{R}}$.

Condition 1 forbids the obvious sampler. Drawing the positive quota from the mixture $\bar{p}_i = \sum_r \omega_r p_{i,r}$ is natural but wrong: its support is dominated by the broadest walk, so the sharpest scale receives one or two scored candidates and its KL term stops carrying ranking information — reintroducing by sampling exactly the collapse that Proposition 1 identifies in the objective. A per-scale quota satisfies the condition; Appendix B gives the allocation we use.

The family is checkable before training. Each result of Section 4 names a quantity that can be computed from the teacher artifact alone, so a degenerate scale family is detectable before the first gradient step:

$$\underbrace{D_{\text{KL}}(p_{i,r} \parallel \mathcal{U}_{\text{supp}(p_{i,r})})}_{\text{sharpness}}, \quad \underbrace{D_{\text{KL}}(p_{i,r} \parallel p_{i,r'})}_{\text{distinctness}}, \quad \underbrace{\text{JS}_\omega(\{p_{i,r}\})}_{\text{floor equation 7}}. \quad (11)$$

Sharpness near 0 means the target is a binary neighbour label carrying no ranking signal; distinctness near 0 means r and r' are duplicates; a floor near 0 certifies the family is degenerate by Corollary 1. These also select τ_g and bound a useful $\mathcal{R}$ without a downstream sweep, and they matter during training as well: by Corollary 1 the loss value alone separates neither a student that has learned the neighbourhood ranking from one whose embeddings have collapsed towards uniform, nor a converged student from an exhausted objective (Appendix B).

Cost. Graph construction is a one-off $O(Nk)$ sparse operation over cached embeddings, and the $r = 0$ target is a lookup in the same cache. Training never touches the teacher: the memory resident during optimisation is the student plus an $N \times d_T$ table of teacher vectors, and the additional runtime cost over a mini-batch relational objective is $|\bar{\mathcal{R}}|$ softmaxes over $|\mathcal{C}_B|$ columns per anchor, negligible next to the encoder forward pass.

6 EXPERIMENTS

The experiments follow the predictions of Section 4 rather than an average: we check that the scale family is non-degenerate before training, report downstream quality, then test Predictions 1–3, each of which specifies what should move *and* what should not.

Protocol. The student is BERT-base (110M, $d_S = 768$). We distil from three teachers of different families and sizes — BGE-M3, Qwen3-Embedding-0.6B and Qwen3-Embedding-4B — to check that the analysis is not an artifact of one teacher’s geometry. Teacher embeddings are computed once over a deduplicated corpus of 13.5k texts and cached; the teacher is not loaded during optimisation. We train 5 epochs with AdamW at 2×10^{-5} , batch size 16, redrawing candidate sets each epoch. Unless stated otherwise $\mathcal{R} = \{1, 2, 4\}$, $k = 50$, $\tau_g = 0.05$; all constants are in Appendix B.

Evaluation and baselines. Nine benchmarks in three families: linear-probe classification F_1 (Bank77, TweetEval, Emotion), pair classification average precision (MRPC, SciTail, WiC), and STS Spearman correlation (SICK-R, STS12, STS-B). The split into *level-sensitive* metrics (pair classification and STS, which score pairs drawn from many different anchors on one common scale) and *probe-based* metrics (linear probes, which fit their own boundary on the embeddings and never compare cosines across anchors) is fixed in advance by Prediction 1 and is the axis along which we read every ablation; three datasets overlap the corpus in domain and are marked separately. Against the undistilled student and unsupervised SimCSE we compare five recent embedding distillation methods (CDM, DSKD, Jasper & Stella, DistillCSE, EMO) and the strongest prior relational method (TALAS). We report SEED (Fang et al., 2021) explicitly as the $\bar{\mathcal{R}} = \{0\}$ member of equation 3: it is the reference point against which a gain is attributable to $\mathcal{R}$ and the temperature bank rather than to relational distillation as such.

6.1 THE SCALE FAMILY IS NON-DEGENERATE

Table 4 in Appendix B.4 reports the build-time diagnostics of Section 5 for the Qwen3-4B teacher. Sharpness equation 11 is bounded away from zero at every scale (0.65, 1.72, 1.53 nats for $r = 1, 2, 4$), so the targets carry ranking information rather than a binary neighbour label. The pairwise divergences are small but non-zero and decrease with r (0.173, 0.127), quantifying the mixing argument of Section 3: beyond a few lazy steps the scales duplicate one another, and this diagnostic — not a downstream sweep — is what bounds a useful $\mathcal{R}$. Finally $JS_\omega = 0.100$ nats: positive, so Corollary 1 does not vacuously certify degeneracy, yet small enough that a tied-readout run would spend most of its loss on a constant. The truncation residual is numerically zero at $r = 1$ and grows to 8.7×10^{-2} at $r = 4$, reported rather than assumed away.

6.2 DOWNSTREAM QUALITY

For the Qwen3-Embedding-4B teacher, HeatBank reaches 78.36 average against 77.60 for TALAS, the strongest prior method. The gain is not uniform, but neither is it as cleanly differential as Prediction 1 asks. The three largest moves are level-sensitive and positive — STS-B (+3.01), STS12 (+2.00), SciTail (+2.72) — yet MRPC (−1.82) and SICK-R (−0.57) move the other way, so the level-sensitive average rises by only +0.89. The probe-based average rises by +0.51, and that figure is carried entirely by Emotion (+3.33) against declines on Bank77 (−0.91) and TweetEval (−0.90). A differential of +0.89 against +0.51 points in the predicted direction and is, at one seed, too small to be evidence for it. We therefore do not read Prediction 1 off this table at all: a comparison against a different method varies the objective, the targets and the optimisation together, whereas the ablations of Section 6.3 vary one component against a fixed baseline and are where the prediction is actually tested. In-domain and out-of-domain averages both improve over TALAS (+2.12 and +0.09), so the effect is not a corpus-overlap artifact, though the out-of-domain margin is small.

The remaining teacher blocks and the SEED row are not yet run and are shown as “—”, which leaves three things unestablished: the HeatBank row is a *single seed*; the comparison to SEED — the ablation separating our contribution from the relational baseline it builds on — is pending; and the claim that the analysis is teacher-independent rests on one teacher.

6.3 TESTING THE ANALYSIS

Each result of Section 4 names a quantity that a single readout leaves free, so each can be tested twice: *directly*, by measuring that quantity inside the model, and *indirectly*, by the downstream damage its absence causes. Table 2 reports the first and Table 3 the second. We weight the direct measurements more heavily. A downstream average is a noisy, many-step consequence of the mechanism a proposition describes, whereas the excess over the floor, the level gap, the off-graph repulsion rate and the anchor gradient share are the propositions themselves written as numbers — and a mechanism that fails to appear there is not rescued by a favourable benchmark average.

The statistic the predictions actually make. Predictions 1 and 2 do not say “the variant is worse”; they say it is worse *on level-sensitive metrics specifically*. The corresponding statistic is the excess damage $D = \Delta_{\text{Probe}} - \Delta_{\text{Level}}$ relative to the full method, and a variant that degrades both families equally has $D \approx 0$ and refutes the prediction as surely as one that degrades neither. D is a difference

Table 1: Downstream embedding quality across nine benchmarks. Datasets marked $\star$ are in-domain with respect to the distillation corpus. Classification is linear-probe macro F_1 , pair classification is average precision, STS is Spearman correlation. “—” = not run; the HeatBank row is a single seed.

	Classification (F_1)			Pair Cls. (AP)			STS (ρ)			Domain		Avg
	Bank77	Tweet	Emot. $\star$	MRPC	SciTail	WiC $\star$	SICK	STS12	STSB $\star$	In	Out	
Student (no distil.)	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
BGE-M3 $\rightarrow$ BERT-base												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
CDM	—	—	—	—	—	—	—	—	—	—	—	—
DSKD	—	—	—	—	—	—	—	—	—	—	—	—
Jasper & Stella	—	—	—	—	—	—	—	—	—	—	—	—
DistillCSE	—	—	—	—	—	—	—	—	—	—	—	—
EMO	—	—	—	—	—	—	—	—	—	—	—	—
TALAS	—	—	—	—	—	—	—	—	—	—	—	—
SEED ($= r=0$ only)	—	—	—	—	—	—	—	—	—	—	—	—
HeatBank	—	—	—	—	—	—	—	—	—	—	—	—
Qwen3-Embedding-0.6B $\rightarrow$ BERT-base												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
CDM	—	—	—	—	—	—	—	—	—	—	—	—
DSKD	—	—	—	—	—	—	—	—	—	—	—	—
Jasper & Stella	—	—	—	—	—	—	—	—	—	—	—	—
DistillCSE	—	—	—	—	—	—	—	—	—	—	—	—
EMO	—	—	—	—	—	—	—	—	—	—	—	—
TALAS	—	—	—	—	—	—	—	—	—	—	—	—
SEED ($= r=0$ only)	—	—	—	—	—	—	—	—	—	—	—	—
HeatBank	—	—	—	—	—	—	—	—	—	—	—	—
Qwen3-Embedding-4B $\rightarrow$ BERT-base												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	<u>71.53</u>	68.65	63.37	71.42	65.82	75.45	72.24
Jasper & Stella	89.71	73.83	65.98	<u>85.33</u>	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	<u>72.61</u>	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	<u>71.47</u>	79.14	76.58
TALAS	92.34	<u>73.74</u>	64.41	85.97	<u>82.66</u>	68.65	78.38	72.27	<u>79.98</u>	71.01	<u>80.89</u>	<u>77.60</u>
SEED ($= r=0$ only)	—	—	—	—	—	—	—	—	—	—	—	—
HeatBank	<u>91.43</u>	72.84	<u>67.74</u>	84.15	85.38	68.65	<u>77.81</u>	74.27	82.99	73.13	80.98	78.36

of two noisy averages, so it is uninterpretable without a scale: we run the full method at three seeds and report its spread in the same units, and count no D that does not exceed it. This is also why we do not read these predictions off Table 1: comparing against a different method varies the objective, the targets and the optimisation at once, whereas every row below changes one component against a fixed baseline.

(A) The collapse. The tied-readout variant holds the target family and the weights fixed and changes only whether the student side varies with r . Here the theory admits a check sharper than any metric comparison: by equation 6 the tied run and a single-scale run on $\bar{p}$ are the *same optimisation problem*, so we compare their per-step losses and gradients directly and report the maximum discrepancy, which should be at numerical precision. A downstream disagreement between those two rows indicts the implementation, not the analysis. The mixture-proportional sampler then shows the same failure entering through the candidate draw rather than the objective: it violates Condition 1 by starving the sharpest scale of scored columns, and should reproduce part of the tied-readout damage even though equation 10 is untouched.

(B) The levels. The level gap of Table 2 is the direct test: if within-anchor agreement with the teacher is high while pooled agreement is much lower, the student has learned the ranking and not the levels, which is Proposition 2(a) verbatim. Two rows then probe how much of the bank is needed. Using two distinct temperatures while keeping $\bar{\mathcal{R}}$ unchanged tests the minimal case of part (b) without also removing scales — a variant that shrank $\bar{\mathcal{R}}$ would confound the size of the bank with the size of the family — and making τ_r decrease with r tests whether only the *distinctness* of the temperatures is load-bearing, as the analysis claims, or whether the monotone assignment we

Table 2: Mechanism measurements: for each result of Section 4, the quantity that result is about, measured inside the model rather than through a downstream proxy. *Excess* is $\mathcal{L} - \text{JS}_\omega$ at convergence, which Corollary 1 forces to ≥ 0 under a single readout and only bounds below under a bank. *Level gap* is the mean within-anchor Spearman correlation between student and teacher cosines minus the same correlation computed on all pairs pooled across anchors; it is exactly the discrepancy the per-anchor shift of Proposition 2(a) creates. *Off-graph repulsion* is the fraction of teacher-near, off-pool pairs whose student cosine decreases over training, which equation 9 drives to 1 absent a full-support target. *Anchor share* is $\|\nabla_\theta \mathcal{L}_{\text{anc}}\| / \|\nabla_\theta \mathcal{L}\|$ at the last epoch. “—” = not run.

Variant	Excess $\mathcal{L} - \text{JS}_\omega$ (Cor. 1)	Level gap (Prop. 2)	Off-graph repulsion (Prop. 3)	Anchor share (Lem. 1)
<i>Predicted</i>				
Single readout	$\rightarrow 0^+$	large	—	—
HeatBank (full)	< 0	small	$\approx$ base rate	—
No full-support target	—	—	$\rightarrow 1$	—
With $\mathcal{L}_{\text{anc}}$	—	unchanged	—	$\rightarrow 0$
<i>Measured</i>				
HeatBank (full)	—	—	—	n/a
Tied readout, $\tau_r \equiv \tau$	—	—	—	n/a
SEED, $\bar{\mathcal{R}} = \{0\}$	n/a	—	—	n/a
Drop $r = 0$ (graph only)	—	—	—	n/a
Full + $\mathcal{L}_{\text{anc}}$, W_a free	—	—	—	—

adopt is doing work the analysis does not account for. Dropping the shared column set restores the per-anchor freedom that part (b) removes only when anchors compete over common columns.

(C) The support. Dropping $r = 0$ leaves no full-support target, so by equation 9 every off-pool candidate is pushed away at a rate independent of its teacher similarity; the off-graph repulsion rate should approach 1, and its downstream shadow should fall in the level-sensitive column. That variant alone is confounded, however: removing $r = 0$ also renormalises ω over the remaining scales and discards the only dense teacher signal. We therefore add a control that keeps ω_0 and the teacher’s mass and changes support alone, restricting $p_{i,0}$ to the pool $\mathcal{P}_i$. If the damage is a support effect this row reproduces most of it; if it does not, what mattered was the dense signal rather than Proposition 3. Replacing $r = 0$ by uniform smoothing completes the dissection, separating *having* full support from *having the teacher’s* mass on it, and should recover only part of the gap.

(D) The pointwise anchor. The question is substitution, not addition, so the load-bearing row adds $\mathcal{L}_{\text{anc}}$ to a student that still lacks the bank: if a pointwise term supplied calibration, tied readout + $\mathcal{L}_{\text{anc}}$ would recover the level-sensitive column, and Lemma 1 says it cannot. Adding it to the full method is the complementary null — worth committing to in advance — and the anchor gradient share measures the mechanism: the term should fit the best linear predictor of t_i from s_i early and then contribute almost nothing. The orthogonal- W_a row is the control that distinguishes the two available explanations: if constraining W_a helps where a free W_a does not, the failure was the $GL(d_S)$ gauge rather than pointwise supervision as such.

7 DISCUSSION

Scope, and why temperatures. Propositions 1–3 say nothing about heat diffusion: they concern matching a family of targets through a softmax readout, and apply wherever such a family arises — multi-teacher, multi-view or multi-layer objectives, curriculum schedules over target sharpness. Other r -dependent student maps would also break the collapse identity — a per-scale projection head, say — but they add parameters discarded at inference and reintroduce the very $GL(d_S)$ freedom that Lemma 1 identifies as the reason pointwise anchoring fails. The temperature bank is the minimal intervention, and by Proposition 2(b) it already supplies calibration.

Table 3: Downstream ablations, grouped by the result each isolates. *Level-sens.* averages the six metrics that score pairs drawn from many different anchors on one common scale (MRPC, SciTail, WiC average precision; SICK-R, STS12, STS-B Spearman ρ), which depend on the per-anchor level Proposition 2(a) leaves free; *Probe* averages the three linear probes (Bank77, TweetEval, Emotion), which fit their own boundary on the embeddings and never compare cosines across anchors. Since Predictions 1–3 are differential, the test statistic is $D = \Delta_{\text{Probe}} - \Delta_{\text{Level}}$, the excess damage a variant does to the level-sensitive family; $D > 0$ supports a prediction, $D \approx 0$ refutes it whether or not the variant is worse overall. *Pred.* fixes the expected sign before the run (++ large, + positive, 0 none). D counts only if it exceeds the seed spread in the reference block. “–” = not run.

Variant	Isolates	Pred.	Level-sens.	Probe	D
<i>Reference points</i>					
HeatBank (full)	—	—	78.88	77.34	—
seed spread, $n = 3$	noise floor for D	—	—	—	—
SEED, $\bar{\mathcal{R}} = \{0\}$	no bank, no diffusion	+	—	—	—
TALAS (strongest prior)	different objective	n/a	77.98	76.83	n/a
<i>(A) Prop. 1: one readout collapses the family onto its mixture</i>					
Tied readout, $\tau_r \equiv \tau$	Pred. 1	++	—	—	—
vs. single scale on $\bar{p}$	Eq. equation 6 is exact	0	—	—	—
Mixture-proportional sampler	Cond. 1: collapse by sampling	+	—	—	—
<i>(B) Prop. 2: one readout leaves the per-anchor level free</i>					
Two distinct τ , $\bar{\mathcal{R}}$ fixed	part (b), minimal case	+	—	—	—
τ_r decreasing in r	distinctness vs. assignment	0	—	—	—
Per-anchor columns, no $\mathcal{C}_B$	levels across anchors	+	—	—	—
<i>(C) Prop. 3: truncated targets repel off the graph</i>					
Drop $r = 0$ (graph only)	Pred. 2	++	—	—	—
$r=0$ restricted to $\mathcal{P}_i$	support alone, ω_0 and mass held	++	—	—	—
$r=0$ replaced by uniform	support without the teacher’s mass	+	—	—	—
<i>(D) Lem. 1: a pointwise anchor cannot supply the missing quantity</i>					
Tied readout + $\mathcal{L}_{\text{anc}}$	can it substitute for the bank?	++	—	—	—
Full + $\mathcal{L}_{\text{anc}}$, W_a free	Pred. 3, a null	0	—	—	—
Full + $\mathcal{L}_{\text{anc}}$, W_a orth.	control: $GL(d_S)$ removed	0	—	—	—

Limitations. The evidence is one teacher and one seed: Table 1 has two teacher blocks unfilled and no variance estimate, so the +0.76 aggregate is a not-yet-contradicted expectation rather than a result. Every prediction in Section 4 is stated but untested; the SEED comparison is the most load-bearing missing cell, and the differential structure of Prediction 1 means a merely-positive ablation would not confirm it. $\bar{\mathcal{R}}$ is bounded above by mixing rather than by choice, so whether a sparser graph widens the usable range is open. And Condition 1 constrains the sampler rather than the objective, so an implementation can satisfy equation 10 while silently violating it.

Conclusion. The difficulty in relational distillation lies in the readout rather than in the targets. One softmax makes a family of targets act only through its mixture, determines the student’s similarities only up to a per-anchor shift, and converts truncated targets into a force that separates pairs the teacher calls close. A bank of per-scale temperatures over a family that includes the zero-step, full-support target removes all three at no parameter cost, and each claim implies a differential prediction — what should move and what should not — so the experiments can falsify the analysis rather than summarise it.

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A DEFERRED STATEMENTS AND PROOFS

Lemma 1 (Gauge). *Let $\mathcal{L}_{\text{rel}}$ depend on the student only through $\{\cos(s_i, s_j)\}_{i,j}$; its invariance group contains $G_{\text{rel}} = \{s \mapsto cRs : c > 0, R \in O(d_S)\}$. For $\mathcal{L}_{\text{anc}} = \frac{1}{|B|} \sum_{i \in B} (1 - \cos(W_a s_i, t_i))$ with $W_a \in \mathbb{R}^{d_T \times d_S}$ free, the substitution $s \mapsto As, W_a \mapsto W_a A^{-1}$ leaves the term exactly invariant for every $A \in GL(d_S)$. Hence $G_{\text{anc}} = GL(d_S) \supsetneq G_{\text{rel}}$, and $\min_{W_a} \mathcal{L}_{\text{anc}}$ is a function of the student space only through its $GL(d_S)$ -orbit.*

Proof. For any invertible A , $W_a A^{-1}(As_i) = W_a s_i$, so every summand is unchanged; the map $W_a \mapsto W_a A^{-1}$ is a bijection of the parameter space, so the minimised value is unchanged as well. Taking $A = cR$ recovers $G_{\text{rel}} \subseteq G_{\text{anc}}$, and any A with distinct singular values lies in $G_{\text{anc}} \setminus G_{\text{rel}}$, which gives strict containment. $\square$

Proof of Proposition 2. (a) is the shift invariance of the softmax: replacing v_i by $v_i + a\mathbf{1}$ multiplies numerator and denominator of equation 5 by $e^{a/\tau}$.

(b) Exact satisfaction $q_i^{\tau_r} = p_{i,r}$ holds iff $v_i/\tau_r = \log p_{i,r} + c_r \mathbf{1}$ for some $c_r \in \mathbb{R}$, i.e. $v_i = \tau_r \log p_{i,r} + \tau_r c_r \mathbf{1}$. Equating two such expressions for $r \neq r'$ gives equation 8. The first constraint already pins v_i up to one scalar; each further r imposes $C - 1$ additional conditions on that single remaining degree of freedom, which for generic targets admit no solution. The minimiser of $\sum_r \omega_r D_{\text{KL}}(p_{i,r} \| q_i^{\tau_r})$ is therefore a compromise, and since the residuals it balances are the vectors $\tau_r \log p_{i,r}$ themselves, its location depends on the absolute values of $\log p_{i,r}$ and not only on the shift-equivalence class of v_i . $\square$

B IMPLEMENTATION DETAILS

This appendix fixes the constants left symbolic in the main text. Nothing here is required by the analysis of Section 4; the choices are reported so that Table 1 is reproducible and so that Condition 1 can be checked rather than assumed.

Algorithm 1 HeatBank training

Require: Corpus $\mathcal{D}$, teacher T , student S_θ , graph size k , scales $\bar{\mathcal{R}} = \{0\} \cup \mathcal{R}$, temperatures $\{\tau_r\}$, weights $\{\omega_r\}$, pool size P , candidate size C

- 1: Cache $t_i = T(x_i)$ for all i ; discard T
- 2: Build W by equation 2; form P and Q as in Section 3
- 3: Precompute pools $\mathcal{P}_i$ and targets $p_{i,r}$, $r \in \mathcal{R}$, from one walk snapshotted at each r ; record residuals $\varepsilon_{i,r}$
- 4: Compute the floor equation 7 and the build-time diagnostics equation 11
- 5: **for** each epoch e **do**
- 6: Redraw $C_i^{(e)} \sim \mu_i$ for every anchor i {Condition 1}
- 7: **for** each mini-batch B **do**
- 8: Encode anchors and candidates with S_θ ; form the shared column set $\mathcal{C}_B$ and scatter targets onto it
- 9: Evaluate $p_{i,0}$ on $\mathcal{C}_B$ from the cached teacher embeddings
- 10: Compute $q_i^{\tau_r}$ for every $r \in \bar{\mathcal{R}}$ and update θ on equation 10
- 11: **end for**
- 12: **end for**
- 13: **return** S_θ

B.1 HYPERPARAMETERS

Symbol	Meaning	Value
k	mutual k NN graph degree cap	50
τ_g	graph kernel temperature, equation 2	0.05
τ_0^{tgt}	teacher temperature of the $r=0$ target, equation 3	0.10
$\mathcal{R}$	diffusion scales	$\{1, 2, 4\}$
(τ_1, τ_2, τ_4)	readout temperatures, diffusion scales	$(0.05, 0.07, 0.10)$
τ_0	readout temperature, zero-step scale	0.10
$(\omega_0, \omega_1, \omega_2, \omega_4)$	scale weights, normalised	$(0.500, 0.286, 0.143, 0.071)$
P	retained pool size per anchor	128
C	candidate set size per anchor	64
—	optimiser / LR / batch / epochs	AdamW / 2×10^{-5} / 16 / 5

The diffusion readout temperatures increase with r , so that broader targets are read out more smoothly. The zero-step readout temperature is set independently rather than by extending that trend: $q_i^{\tau_0}$ ranges over the entire candidate set rather than a k NN neighbourhood, where inheriting τ_g would place nearly all mass on the nearest column, and it is paired with a target whose own temperature is τ_0^{tgt} . Proposition 2(b) requires only that at least two readout temperatures differ, which the bank satisfies. The graph temperature τ_g is selected by the sharpness diagnostic of equation 11 rather than by a downstream sweep: teacher cosines within an anchor’s top- k neighbourhood occupy a narrow band, so a τ_g large relative to that band flattens every row of P towards uniform on its support, which sharpness detects offline.

The weights follow from normalising the raw diffusion weights $(1, \frac{1}{2}, \frac{1}{4})$ to sum to one and then admitting $r = 0$ with unit relative weight, so the zero-step scale carries half the total. We record this explicitly because it is easy to introduce a scale whose effective weight is far from what a list of raw coefficients suggests, and because ω enters both the objective equation 10 and the floor equation 7 — a floor computed under different weights than the loss is not comparable to it.

B.2 THE CANDIDATE SAMPLER

Condition 1 is realised by a fixed quota allocation of the $C = 64$ slots:

- 32 diffusion slots, split evenly across $\mathcal{R}$ (not drawn from the mixture $\bar{p}_i$), so each scale is guaranteed scored columns from its own support;
- 24 teacher hard negatives, drawn from the 200 nearest teacher neighbours that lie outside the graph pool;
- 8 uniform random negatives.

Within a scale’s quota, the top 2 columns by diffusion mass are taken deterministically and the remainder are drawn without replacement by Gumbel top- k over $\log(P_r)_{ij}$, which samples proportionally to diffusion mass while keeping the draw vectorised. Sets are redrawn once per epoch, for a reason independent of Condition 1: a frozen candidate set defines a fixed finite collection of comparisons, so once the student orders them correctly the gradient vanishes even though $p_{i,r}$ still describes unlearned structure. If mutuality isolates an anchor — 2.0% of anchors on our corpus — its pool falls back to the teacher’s directed top- k neighbours; these anchors still receive a well-defined $r = 0$ target, which is one practical consequence of Proposition 3’s remedy.

B.3 CORPUS PREPARATION AND TARGETS

Exact-duplicate texts are removed before graph construction (13.5k anchors remain): they have $\cos = 1$ by construction, so their logits sit at the ceiling with vanishing gradient while still absorbing teacher mass from candidates that could still move. Diffusion targets are precomputed once by a single sparse walk snapshot at each $r \in \mathcal{R}$, retaining the top $P = 128$ entries per row and recording the truncation residual $\varepsilon_{i,r}$ of equation 4; the $r = 0$ target is evaluated on the fly from cached teacher vectors, so it is exact on $\mathcal{C}_B$ and requires no storage. All targets are renormalised on the scored column set after masking the anchor’s own index.

B.4 BUILD-TIME DIAGNOSTICS

B.5 LOGGED QUANTITIES

Per epoch we log: total loss and per-scale KL; the excess $\mathcal{L} - \text{JS}_\omega$; student entropy relative to $\log |\mathcal{C}_B|$ per scale; student mass on the teacher’s top-8 candidates; mean and 90th-percentile truncation residual; and the number of steps skipped for non-finite gradients. The last is reported because a silently skipped step is indistinguishable from a completed one in the loss curve. Checkpoint selection uses a held-out validation metric, not the training objective: by Corollary 1 the training loss contains a large anchor-dependent constant, so its across-epoch differences are not a reliable ordering of model quality.

Table 4: Build-time diagnostics of the scale family equation 3 for the Qwen3-Embedding-4B teacher ($k = 50, \tau_g = 0.05$), averaged over anchors and evaluated on the retained pool. Sharpness is the first quantity in equation 11, in nats; a value near 0 would indicate a binary neighbour label. Residual is $\varepsilon_{i,r}$ of equation 4. The zero-step scale is dense on C_i by construction and is reported separately in the text.

Scale	$ \text{supp}(p_{i,r}) $	Sharpness equation 11	Top-1 mass	Residual $\varepsilon_{i,r}$
$r = 1$	20.5	0.649	0.401	1.4×10^{-17}
$r = 2$	101.8	1.719	0.333	2.4×10^{-2}
$r = 4$	124.9	1.528	0.257	8.7×10^{-2}

Scale distinctness: $D_{\text{KL}}(p_{i,1} \| p_{i,2}) = 0.173$, $D_{\text{KL}}(p_{i,2} \| p_{i,4}) = 0.127$
Irreducible floor equation 7: $\text{JS}_\omega = 0.100$ (mean), 0.191 (90th percentile)
Graph: mean degree 20.5; 2.0% of anchors isolated by mutuality and restored by fallback